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Improved 3D reconstruction pipeline for enhancing the quality of 3D model generation from multi-view images

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In this research, we slightly modify the conventional pipeline such that it (1) sets a minimum triangulation angle of 3°, (2) reduces (and possibly minimizes) overall re-projection error by simultaneously optimizing all camera poses and 3D points in the bundle adjustment step, and (3) uses a tiling buffer size of 1024×1024, to generate, from a set of 2D images, a detailed 3D models of complex objects. We systematically showcase the versatility of this improved photogrammetric pipeline by applying it to three datasets - a proprietary dataset (money plant) and two publicly available datasets (a statue and an old computer). To the best of our knowledge, while previous research has employed photogrammetry techniques for 3D model generation, none has systematically defined the 3d reconstruction pipeline, with a focus on underlying mathematical concepts, Additionally, there has been a lack of evaluation of how the quality of input 2D images impacts quality of the resulting 3D models. Our study evaluates the entropy and image quality (using BRISQUE scores) of the 3D model generated in relation to the image quality of the set of input 2D images, providing insights into the reliability of the proposed photogrammetric approach. Despite utilizing a low-quality image dataset as input, we demonstrated that we achieved a high-quality 3D model. Furthermore, we observed less differences in image quality when comparing the 3D model with the original 2D images. Overall, our research contributes to both theoretical understanding and practical application of photogrammetry for 3D reconstruction as experimented on three distinct image datasets. As a future research scope, we provide suggestions for possible parallelization of some of the steps of the 3D reconstruction pipeline.

Keywords Photogrammetry, 3D model, Point cloud, Image quality

Various methods exist for generating 3D models, each tailored to specific domain applications and data. These methods are used in Photogrammetry, Structure from Motion (SfM), Light Detection and Ranging (LiDAR) scanning, Medical imaging, Computer-aided Design (CAD), Depth sensing, and Procedural modeling. Computer vision techniques, such as SfM[1], allow for the retrieval of three-dimensional models from two-dimensional images or video. These methods can reconstruct three-dimensional geometry and produce three-dimensional models using visual input. Photogrammetry[2] utilizes photographs for 3D reconstruction, while 3D scanning employs lasers or structured light. LiDAR scanning[3] measures distances with laser light, and structured light scanning projects patterns onto objects. Medical imaging techniques, such as CT/MRI[4], captures internal structures in 3D. Procedural modeling generates models algorithmically, CAD involves manual creation, and volumetric capture records the space around objects. Point cloud fusion combines data from multiple sources, and depth-sensing cameras use sensors for 3D representation[5]. The choice of the method depends on factors specific to the application.

With the advent of digital data processing, photogrammetry has emerged as an effective substitute for the traditional approaches in 3D reconstruction. Furthermore, it is a cost-effective option due to its affordable investment and equipment requirements, making it suitable for direct field use. Photogrammetry may generate very realistic outcomes by (1) capturing geometric data and (2) projecting the original photos onto the 3D model to produce a texture effect. The 3D models are useful for study, analysis, reconstruction, and documentation[2].

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It is crucial to acknowledge that achieving excellent 3D-model reconstruction frequently necessitates meticulous preparation, appropriate image gathering, and thoughtful considerations regarding the attributes of the scene or object being reconstructed. Users should possess a thorough understanding of the fundamentals of photogrammetry to attain optimal outcomes.

Here, we use a modified photogrammetry pipeline to generate a 3D model using our dataset captured with a mobile phone camera. The modified approach, where we have made slight adjustments to the conventional photogrammetry pipeline, is detailed in Section "Methodology and experimental setup". These modifications incorporate bundle adjustment to optimize point locations and adapt parameters in essential steps like triangulation and camera initialization, utilizing the default calibration method of smartphones. Additionally, we examine publicly accessible datasets for the same purpose and thoroughly explain each individual step involved in generating a 3D model, including the relevant mathematical concepts. We also evaluate the generated 3D model using three different aspects: 1). Visual perspective, 2). Image entropy comparison, and 3). No-reference image quality metric BRISQUE. We also analyze the time taken by each step of the pipeline.

Further, the content of the paper has been organized as follows: In Section "Related work", a comprehensive literature review is conducted on the utilization of photogrammetry for 3D model generation. Section "Theory and concepts" delves into the pertinent background theory essential for our research. Section "Methodology and experimental setup" outlines the methodology and experimental setup employed in our work. Sections "Steps for 3D model generation" and "Methodology and experimental setup" meticulously detail the steps to constructing a 3D model through the photogrammetry pipeline. Evaluation methods are elucidated in Section "3D model evaluation", while Section "Conclusions" encapsulates conclusions and scope for future research.

Related work

The research done by Jasinska A et al.[6] assesses smartphone photogrammetry; they test the "Structure from Motion – Multiview Stereo" (SfM-MVS) procedure on 14 smartphones. It reveals lower stability compared to DSLR cameras, recommending stability tests before metric measurements. External calibration improves 3D model geometry, cautioning against premature use of the term "smartphone photogrammetry" due to limited accuracy studies. The study by Jasinka et al. proposes a separate calibration process. The study delves into the intricacies of smartphone-generated images, but an in-depth evaluation of the resultant 3D models in relation to the quality of the original input images is not covered[6]. Marion L. et al.[7] developed a floral photogrammetry protocol using a DSLR camera, turntable, and portable light box to reconstruct 3D flower models from 2D images rapidly and accurately. This cost-effective method captures all visible parts of flowers while maintaining color information. Demonstrating its application on 18 Gesneriaceae species, the study positions photogrammetry as an affordable alternative to micro-CT, potentially revolutionizing studies of floral evolution and ecology by providing access to 3D morphological data[7].

The work done by Gallego J et al.[8] presents a thorough approach for automatically reconstructing 3D object models from frontal RGB photos, with a specific emphasis on guitars as the main example. Their approach employs convolutional neural networks (CNNs) for object detection and segmentation, followed by 3D reconstruction using depth maps and a fitted 3D template, verified using actual photos and Shape Net renders. Incorporating shape and texture enhances the field of realistic 3D object reconstruction. It may be applied to generating virtual reality content and is adaptable to various types of objects[8].

Oliveira A. et al.[9] demonstrate the utilization of 3D stereoscopic techniques for neuroanatomy education, using photogrammetry to create realistic 3D models. Blender software combines and sets up these models for stereoscopic projections, offering depth perception and 360° visualization. The reproducible and interactive technique complements traditional 3D models and proves valuable when access to laboratory-based learning is limited[9]. Le Mouelic S. et al.[10] utilize scanned digital photos taken by Apollo 17 astronauts during their 1972 Moon mission to create a comprehensive 3D model of the Taurus Littrow Valley's Station 6. Using SfM photogrammetry, they reconstructed boulders and rock samples, providing insights into size, orientation, and preservation. The 3D models are accessible online and integrated into virtual reality for immersive exploration, showcasing the potential of 3D photogrammetric reconstructions for scientific and educational purposes in future lunar missions[10]. Liu W. et al.[11] has proposed deep learning-based 3D model generation method and applied the same on their proposed multi-source 3D dataset. The proposed MS3DQE-Net integrates 3D meshes and 3D point clouds as input data, amalgamating the acquired features to yield a more comprehensive representation of the shape[11].

A wide range of software solutions both open-source and commercial are available for 3D reconstruction tasks. Open-source platforms such as Meshroom and MicMac etc. provide accessible tools for photogrammetry-based model generation, while commercial alternatives like Agisoft Metashape and Pix4Dmapper etc. offer advanced features and professional-grade outputs. Several comparative studies in recent research studies have evaluated the performance of these tools across various domains.

Enesi and Kuqi (2023) conducted a performance comparison between Meshroom and Agisoft, highlighting the strengths and limitations of each in terms of processing efficiency and model accuracy[12]. Similarly, Lau Bik Sing et al. (2022) performed a comparative analysis between AliceVision Meshroom and Pix4Dmapper for generating three-dimensional photogrammetric models, focusing on visual fidelity and usability[13]. In a broader evaluation, Saif Aati et al. (2019) assessed six photogrammetry software packages—Agisoft PhotoScan, Pix4DMapper Pro, MicMac, RealityCapture, ReCap Photo, and ContextCapture[14]. Their study examined the capabilities of these platforms in automatically processing image datasets to generate dense point clouds, particularly within industrial and archaeological contexts.

These studies provide a foundation for understanding the current capabilities of both open-source and commercial photogrammetry software and underscore the importance of selecting tools based on application-specific requirements such as processing speed, cost, accuracy, and ease of use.

We have conducted a comparative analysis of five software solutions in terms of available features, including two open-source tools (Meshroom and MicMac) and three commercial platforms (Agisoft PhotoScan, Pix4DMapper, and ContextCapture), as summarized in Table 1 below.

In this comparison, Meshroom stands out because it's open-source, transparent, and lets users add custom features, making it ideal for students and researchers who want to explore and experiment. In contrast, Agisoft Metashape and Pix4Dmapper are popular commercial tools that deliver high-quality results but come with costs and don't allow much flexibility to modify how they work, which limits their use for learning or innovation. Bentley ContextCapture is a powerful, enterprise-level software designed for large and detailed projects but is expensive and more complex to operate. Although commercial software often produces smoother and more polished outcomes, their closed systems restrict customization, making them less suitable for educational or research purposes.

Several studies have evaluated photogrammetry software commonly used in archaeology. This paper focuses on open-access, close-range photogrammetry tools for 3D reconstruction, primarily for educational use.

Also, it has been observed that there is less research[8] which compares the quality of the 3D-Model with respect to the quality of the input image-set.

Theory and concepts

In this section, we will discuss the essential theoretical foundations and concepts for a comprehensive understanding of the methodologies and approaches employed in our study.

A. ***Photogrammetry***.

Photogrammetry, as a field, uses photography and various techniques to measure and document precise surface points, creating detailed 3D models of objects or environments. This field includes various techniques and methodologies, such as Structure from Motion (SfM) and Multi-View Stereo (MVS), which are used to capture and process photographic data to produce (accurate) 3D reconstructions.

It can generate intricate 3D models with great precision based on real-life items or settings. This process entails capturing many images of an object from various perspectives and utilizing software to create a comprehensive 3D model with intricate details[2].

Structure-from-motion photogrammetry can be employed to generate 3D models[15]. Photogrammetry includes techniques to measure real-world objects that relies on a conventional camera. The camera captures a sequence of photographs from multiple perspectives and situations. Another frequent approach is to employ an array of synchronized cameras to capture a photograph simultaneously[16]. The photographs are subsequently analyzed using photogrammetry software to produce the three-dimensional model. The software uses algorithms to identify and compare shared characteristic points within the photos. Due to the varying positions of the captured photographs, the feature point undergoes a change in the other image, enabling the software to ascertain the depth of the feature point. This phenomenon is akin to how people can perceive depth through parallax motion, such as when observing the view outside a train window while it is in motion. Proximate objects, such as signal installations at the rails, will traverse the viewer's field of view rapidly, while distant trees may appear to move at a slower pace[1,15].

Feature	Meshroom	MicMac	Agisoft Metashape	Pix4Dmapper	Bentley ContextCapture
License Type	Open-source (free)	Open-source (free)	Commercial (paid)	Commercial (paid)	Commercial (paid)
Pipeline Transparency	Fully transparent and modifiable	Fully transparent, modifiable	Closed-source, limited customization	Closed-source, limited customization	Closed-source, limited customization
Ease of Use	User-friendly GUI, intuitive	Command-line interface, steep learning curve	User-friendly GUI, professional-grade	User-friendly GUI, professional-grade	Professional GUI, complex workflows
Algorithmic Flexibility	High (custom code integration possible)	High (customizable via code)	Low (black-box processing)	Low (black-box processing)	Low (black-box processing)
Installation	Simple (Windows, Linux)	Complex, manual setup, Docker support	Straightforward	Straightforward	Complex (enterprise setup)
Processing Speed	GPU-accelerated, faster on large datasets	CPU-based, slower on large datasets	GPU & CPU optimized	GPU & CPU optimized	High-performance, scalable
Suitability for Education	Excellent (free, modifiable, easy to learn)	Moderate (free but complex to learn)	Moderate (costly, less modifiable)	Moderate (costly, less modifiable)	Low (complex, costly)
Output Quality	Good (depends on customization)	Very high (accurate, professional)	High (industry standard)	High (industry standard)	Very high (enterprise scale)
Support and Community	Strong open-source community	Active open-source community	Professional technical support	Professional technical support	Enterprise-level support
Platform Availability	Windows, Linux	Windows, Linux	Windows, macOS	Windows, macOS	Windows
Trial Limitations	None	None	Limited features, unstable in trial	Limited features, license required	Limited features, license required
Custom Feature Implementation	Possible (open-source, flexible)	Possible (open-source, flexible)	Not possible	Not possible	Not possible

Table 1. Comparative analysis of 3D reconstruction Software.

B. ***Structure from motion***.

Structure from Motion (SfM), is a computer vision methodology employed in photogrammetry to reconstruct the 3D structure of a scene or object. This is achieved by analyzing a set of 2D photographs captured from various perspectives[1]. The objective of SfM is to reconstruct the precise spatial configuration of points in the surrounding environment and the trajectory of the camera that recorded those points.

By studying the movement and corresponding characteristics, it is possible to construct a three-dimensional model of the same object. SfM is widely used in various fields, including 3D modeling, virtual reality, augmented reality, and robotics. It forms the foundation of many photogrammetric and computer vision systems, allowing for the creation of detailed and accurate 3D models from images captured by cameras[17,18].

SfM uses different types of algorithms for feature extraction, like SIFT[19] and ORB (Oriented FAST and rotated BRIEF). SIFT is used for our experiment. Scale-Invariant Feature Transform (SIFT) algorithm is a computer vision technique used to identify and characterize local features in images[17]. The SIFT method can detect and describe key features present in numerous images, irrespective of their scale. This characteristic is extensively utilized in the field of image processing. The SIFT algorithm involves the generation of the Difference of Gaussians (DoG) Space. The SIFT key points of objects are initially collected from a collection of reference images[1] and then saved in a database. In order to identify an object in a fresh image, each feature of the new image is compared separately with images in a database.

Methodology and experimental setup

The experimental setup utilized an HP Zbook Power 15.6-inch G9 mobile workstation PC equipped with DDR5 32GB RAM and NVIDIA RTX A4000 GPU. To produce a 3D model using photogrammetry software, an NVIDIA GPU card with a CUDA compute capability greater than or equal to 3.0 is essential for generating dense, high-quality mesh—capabilities that the Zbook encompasses entirely.

In addition, we have utilized an Apple iPhone equipped with a high-quality camera to capture images of money-plant. Our research employed diverse datasets to construct 3D models using our developed pipeline, as explained in Sections V and VI. The 3D models generated using the given image datasets are compared using the method and three perspectives described in Section VII.

Specifically, we performed our experiments on three datasets, one proprietary dataset and two publicly available datasets. These datasets are listed below:

1) Proprietary dataset of 21 images (Money-plant dataset), colored, JPG.
2) Public dataset of first 40 images of 54 images (Old Computer), colored, JPG[20].
3) Public dataset of 11 images (Statue), colored, JPG[20].

Steps for 3D model generation

A. ***Image acquisition***.

Money plant images (as shown in Fig. 1) were captured using an Apple iPhone 14 pro smartphone camera. Configured the main resolution camera to ProRAW Max (up to 48MP) and personalized the photographic style to a standard, balanced, and true to life (as available options in the mobile). The camera captured multiple photographs of the object by rotating at different angles. A collection of photos is generated for the subsequent stage (15–20 degrees in each rotation clockwise). Figure 1 illustrates images of object captured from 21 distinct viewpoints.

B. ***Pre-processing***.

Fig. 1. Money plant dataset.

During this stage, every image in the dataset underwent pre-processing. Blurry and noisy images are initially detected and subsequently filtered. The images captured with an iPhone were initially in the HEIC format. We converted them into the.

more widely compatible JPG format and decreased their resolution because public dataset images are also in JPG format.

C. ***Photogrammetric reconstruction***.

Photogrammetric reconstruction is the process of creating three-dimensional models, as explained in Section VI.

3D object modeling pipeline

The method we propose uses a photogrammetry pipeline to produce three-dimensional models of tangible objects using a collection of two-dimensional pictures. Figure 2 depicts the procedural sequence for this module.

We used the photogrammetry software Meshroom (2023.1.0) to produce 3D models because it is open-source and compatible with various image formats. It is worth noting that other software solutions are available for this purpose, and customized programs can be created to carry out the necessary processes without relying on specialist software. The input of the pipeline was a compilation of pre-processed images, and the output was a three-dimensional model. The pipeline comprises the subsequent significant stages, as explained below.

A. ***Camera initialization***.

In the camera initialization phase of our experiment, key intrinsic parameters were as follows:

1) Focal Length: 57.94 mm.
2) Pixel Ratio: 1 (Locked).
3) Camera Type: Radial3.
4) Image Dimensions: 1024 × 1024.
5) Sensor Dimensions: 36 × 27.

We employed a Distortion Initialization Mode to address lens distortions, and the specific distortion parameters were carefully determined for optimal correction. These parameters lay the foundation for precise camera calibration, ensuring our experiment's accuracy in subsequent 3D reconstruction stages.

B. ***Feature extraction***.

Feature extraction methods like SIFT or ORB can detect edges, corners, or critical points. In this case, we utilize SIFT.

These features were utilized as identifiable markers crucial for subsequent analysis, allowing for effective tracing and matching across many images.

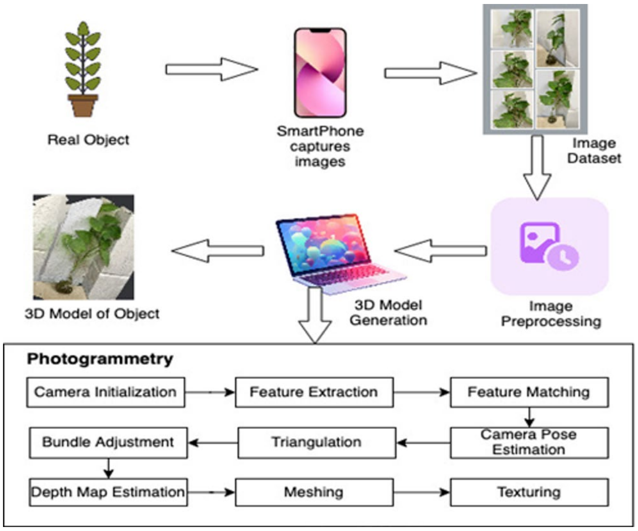

Fig. 2. Photogrammetry architecture. Figure 2 outlines the main stages of our photogrammetry pipeline. To improve texture quality, we set the tiling buffer size to 1024 × 1024 during the camera initialization step. In the bundle adjustment step, we focused on minimizing the reprojection error to achieve more accurate results. Finally, in the triangulation step, we applied a minimum triangulation angle of 3 degrees to ensure better geometric consistency.

Let $I_{i(x,y)}$ denote the intensity value at pixel coordinates (x, y) in image i. The set of features derived from the picture i is denoted as F_i i.e. F_i= Extract_Features (I_i).

C. ***Feature matching***.

Subsequently, related characteristics across several photos are identified and aligned. The pipeline utilizes the identification of shared features across multiple photos to generate correspondence that reflects identical spots in the 3D scene. Given two images $I_i$and I_j, match features F_i and F_j based on descriptors. Let $M(F_i, F_j)$ represent the set of matched features between $F_i$and F_j.

The results of the feature matched progress to the subsequent step, SfM, which is responsible for generating the 3D model. Below is a concise explanation of the functioning of the SfM algorithm in the sub-sections D, E, and F below.

D. ***Camera pose estimation***.

For each photograph in the sequence, SfM predicts the camera poses. It determines the exact location and orientation of the camera when the picture is taken[21,22].

The pair (R_i, T_i) represents the camera pose of the image i, where R_i = rotation matrix (rotation about the intrinsic axes of the camera's coordinate system) and T_i = translation vector.

By utilizing correspondences among matched features (i.e. $M(F_i, F_j)$), we can estimate the essential matrix E_{ij}(when the cameras are calibrated and the fundamental matrix, F_{ij}, has been derived already) or the fundamental matrix F_{ij} (when cameras are not calibrated and the essential matrix, E_{ij}, has been derived already)[22] as follows:

$$F_{ij} = K_j^{-1} \cdot E_{ij} \cdot K_i \quad (1)$$

$$E_{ij} = K_j^T \cdot F_{ij} \cdot K_i^{-1} \quad (2)$$

where $K_i$and $K_j$are the intrinsic matrices of the cameras corresponding to views i and j. We have calculated E_{ij} using Eq. (2) above.

We can also compute the fundamental matrix F using Singular Value Decomposition (SVD). The fundamental matrix F must satisfy the rank-2 constraint, which requires one of its singular values to be zero. This constraint ensures that F correctly represents the epipolar geometry between two views. To enforce this, we decompose the estimated fundamental matrix F into three matrices: U, D, and V^T, using SVD. Here, $F = UDV^T$, where U is a 3×3 orthogonal matrix, D is a 3×3 diagonal matrix containing non-negative real numbers (the singular values) along its diagonal, and V^T, is the transpose of a 3×3 orthogonal matrix V.

R and T matrices can be computed using singular value decomposition (SVD) once we get matrix E.

The full-camera pose $[R_i \mid T_i]$ is obtained by combining the rotation and translation.

$$[R_i|T_i] = [R_j|T_j] \cdot [R_{ij}|T_{ij}] \quad (3)$$

The above expression represents the concatenation (denoted by |) or combination of 3×3 rotation matrix R_i and 1×3 translation vector T_i into a single transformation matrix. The camera pose matrix $[R \mid T]$ is typically a 3×4 matrix that combines rotation and translation components. When multiplying two 3×4 pose matrices, as well as the rotation and translation elements, must be handled separately. For example, given two camera pose matrices $[R1 \mid T1] = [R1\, T1]$ and $[R2 \mid T2] = [R2\, T2]$, each 3×4 matrix can be converted into a 4×4 transformation matrix in homogeneous coordinate system as:

$$T_1 = \begin{matrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \\ 0 & 0 & 0 & 1 \end{matrix} \quad (4)$$

The top left 3 × 3 submatrix in Eq. 4 is the matrix denoting the composition of rotation transformation, and the fourth column is due to translation transformation.

The transformation matrix T_2 is similarly constructed. These two 4×4 transformation matrices in homogeneous coordinate system can then be multiplied: $T = T_1 \times T_2$.

From the resulting 4×4 matrix T, the rotation matrix and translation vector can be extracted to form the final 3×4 pose matrix $[R \mid T]$.

Using this transformation, the cameras are aligned in a single coordinate system.

When the cameras are not calibrated, the camera pose equations include the calibration matrices $K_i$and K_j.

$$x_i = K_i \cdot [R_i|T_i] \cdot X \quad (5)$$

$$x_j = K_j \cdot [R_j|T_j] \cdot X \quad (6)$$

where, $x_i$and $x_j$are the 2D coordinates of a point X in two different camera views.

E. ***Triangulation***.

Triangulation is used to determine the 3D coordinates of the matched features. By analyzing the known camera poses and the positions of the features in multiple views, SfM can estimate the 3D location of each point in the scene. Triangulate the 3D coordinates *P* of a point using the perspective projection equation:

$$P = triangulate\,(x_i, x_j, R_i, T_i, R_j, T_j, K_i, K_j) \tag{7}$$

where $x_i,\ x_j$ are as specified in the description of Eqs. (5) and (6), R_i, T_i, R_j and T_j are camera poses, and K_i and $K_j$are camera calibration matrices.

A minimum of three observations for triangulation is specified to reduce the point cloud's noise effectively. Additionally, we ensured a minimum triangulation angle of 3° to achieve optimal results. Afterward, the Incremental Refinement algorithm[21] is applied iteratively to refine the camera poses and 3D points by considering additional images in the sequence. This refinement helps improve the accuracy of the reconstruction. Refine the camera poses and 3D points iteratively using optimization techniques to minimize the re-projection error expressed as:

$$E_r = min_{\{R_i,T_i\},\{P_k\}} \sum\nolimits_{i,k} dist(x_{ik}, proj\,(P_k, R_i, T_i, K_i)) \tag{8}$$

where, $x_{ik}$is the observed 2D position of the points, P_k is the 3D point coordinates, $dist(.,.)$ computes the distance or error matrix between the two given arguments and $proj(.,.,.,.)$ computes the projection of the point in the given image for the camera pose and 3D point. The operation $min(\ldots)$ minimizes the expression concerning its subscripted parameters.

F. ***Bundle readjustment***.

Bundle adjustment is a key optimization step in SfM. It optimizes the 3D structure (point locations), and camera poses to minimize the difference between the observed image points and their projections in 3D space.

Simultaneously optimize all camera poses $\{R_i, T_i\}$ and 3D points $\{P_k\}$ to minimize the overall re-projection error, E_r:

$$E_r = min_{\{R_i,T_i\},\{P_k\}} \sum\nolimits_{i,k} dist(x_{ik}, proj\,(P_k, R_i, T_i, K_i))^2 \tag{9}$$

G. ***Dense 3d point cloud reconstruction***.

Here, the Multi-View Stereo (MVS)[6] technique produces a dense 3D point cloud and subsequently constructs an intricate 3D model. After obtaining the sparse point cloud in the previous steps, the MVS algorithm works to densely sample the entire scene or object. The idea is to create a dense 3D point cloud by estimating the 3D coordinates of points for every pixel in every image. For parallel setup of stereo camera epipolar lines[21,22] align to horizontal lines, and estimating depth is simple. In another case, when the stereo camera setup is non-parallel and convergent, then we perform image rectification as described by Loop and Zhang[23].

1) Depth Map Estimation.

For each image after rectification, the depth map is estimated. The depth map assigns a depth value to each pixel in the image, representing the distance of the corresponding 3D point from the camera. The depth map $Z_i(x, y)$, i.e. depth value to pixel (x, y) in image i, for each image is estimated using the camera projection equation:

$$D_i(x, y) = f \cdot\ T_i?Z_i(x, y) \tag{10}$$

where *f* is the camera's focal length, T_i is the baseline between stereo cameras and $D_i(x, y)$ is the disparity at the pixel (x, y) in image i. Though Eq. (10) gives more accurate estimates for parallel camera setup, after rectification of the images, the distortion introduced is much less than that of directly applying Eq. (10) for depth estimation[20]. Here, the tiling buffer size is $1024\ \times\ 1024$. Tiling splits the computation into fixed buffers to best fit the GPU.

2) Surface Reconstruction.

Using the estimated depth maps and camera poses, triangulate the 3D coordinates of each pixel in each image to reconstruct the surfaces in 3D space. This step involves refining the geometry and ensuring consistency between multiple views. For each pixel (x, y) in each image, triangulate to obtain the 3D coordinates (X_i, Y_i, Z_i):

$$X_i = (x - c_x)?f \cdot\ Z_i \tag{11}$$

$$Y_i = (y - c_y)?f \cdot\ Z_i \tag{12}$$

$$Z_i = D_i(x, y) \tag{13}$$

where *f* is the camera's focal length, Z_i represents the depth of the 3D point I, and $(c_x,\ c_y)$ are the image center coordinates.

Combine the depth maps from all images to create a fused depth map representing the 3D structure of the entire scene. Fuse the depth maps from multiple images to create a fused depth map, $D_f(x,y)$.

$$D_f(x,y) = min_i\left(D_i\left(x,y\right)\right) \tag{14}$$

Using the fused depth map, merge the individual point clouds obtained from different images into a single dense 3D point cloud. This fusion process aims to integrate information from multiple views to produce a more complete and accurate scene representation. Triangulate the 3D coordinates $(X_f,\ Y_f,\ Z_f)$ using the fused depth map:

$$X_f = \left(\left(x - c_x\right)/f\right) \cdot\ Z_f \tag{15}$$

$$Y_f = \left(\left(y - c_y\right)?f\right) \cdot\ Z_f \tag{16}$$

$$Z_f = D_f(x,y) \tag{17}$$

The output of this step is a dense 3D point cloud that densely samples the entire scene or object, providing detailed geometric information.

H. ***Mapping and texturing***.

Once the dense point cloud is reconstructed, it can be further processed to generate a mesh representation of the 3D surface. Mesh generation[24] involves connecting the points in the point cloud to form triangles, creating a surface mesh that approximates the scene's geometry. The vertices of the mesh can be represented by the 3D coordinates $(X_f,\ Y_f,\ Z_f)$.

Texture mapping[18,24] is being performed to enhance the reconstructed model's visual appearance. This involves projecting the colors from the images onto the reconstructed geometry. The 3D model obtained from these steps is shown in Fig. 3a and b, and 3c visually showcase the resulting 3D models generated from two publicly available datasets (Old Computer[20] and Statue[20]. Furthermore, once the 3D model is generated, multiple views of the model can be presented, providing a comprehensive and immersive 3D experience that provides a better understanding and analysis.

For a clearer comparison between the standard photogrammetric pipeline with default settings and our enhanced approach, please refer to Figs. 4.

As shown in Fig. 4, the 3D models, as shown in the top row, generated using our proposed pipeline demonstrate improved visual quality relative to the corresponding models, as shown in the bottom row in Fig. 4, produced by the standard photogrammetric pipeline with default settings. Additionally, to evaluate image quality, the comparison of the BRISQUE scores has been performed between the corresponding 3D models shown in Fig. 4. The BRISQUE score comparison (i.e. lower the BRISQUE score, better the image quality and vice versa[28] further confirms that the models generated by proposed pipeline deliver better performance in terms of image quality.

3D model evaluation

In the comprehensive evaluation of the 3D model generated through our experiment, we employed three distinct perspectives: 1). Visual Perspective, 2). Entropy (based on image histogram)[25], and 3). Blind/Reference less Image Spatial Quality Evaluator (BRISQUE) no-reference image quality metric[26], a holistic measure of image quality to assess and validate its quality. The no-reference image quality assessment[27,28] metrics like NIQE[28] and PIQE[28] have not been used because NIQE measures deviations from statistical regularities in natural images[27], and PIQE in most of the cases, performs similarly to NIQE[28]. We applied the identical evaluation method to the above-mentioned three datasets, and the statistics of the entropy (of image histogram) and no-reference BRISQUE quality scores are presented in Tables 2 and 3. The discussions with respect to these are detailed below.

A. ***Visual perspective***.

We observed the 3D model through meticulous visual inspection and perceptually determined its fidelity. The qualitative assessment indicates that the generated 3D model exhibits high visual accuracy, showcasing notable attention to detail and overall quality. Readers are advised to see Fig. 3a and b, and 3c in color.

B. ***Entropy comparison***.

We utilized entropy as a measure to quantitatively analyze the information content of both the 2D images in our input dataset and the corresponding 3D model images. The average and maximum entropy of our original 2D image dataset was found to be 5.07 and 5.27, respectively. This implies that the maximum achievable entropy is 5.27. In contrast, the entropy of the image generated from the 3D model was measured at 4.72. The observed difference between the 2D image max entropy and the 3D image of approximately 0.55 suggests a favorable outcome, emphasizing that the generated 3D model effectively captures and retains the essential information content present in the input images. As we can observe from Table 3, the 3D model image for all three datasets (Money plant, OC, statue) has entropy less than the maximum entropy in the corresponding dataset images' entropies. Further, as explained below, the image quality scores are also not good.

C. ***Brisque score comparison***.

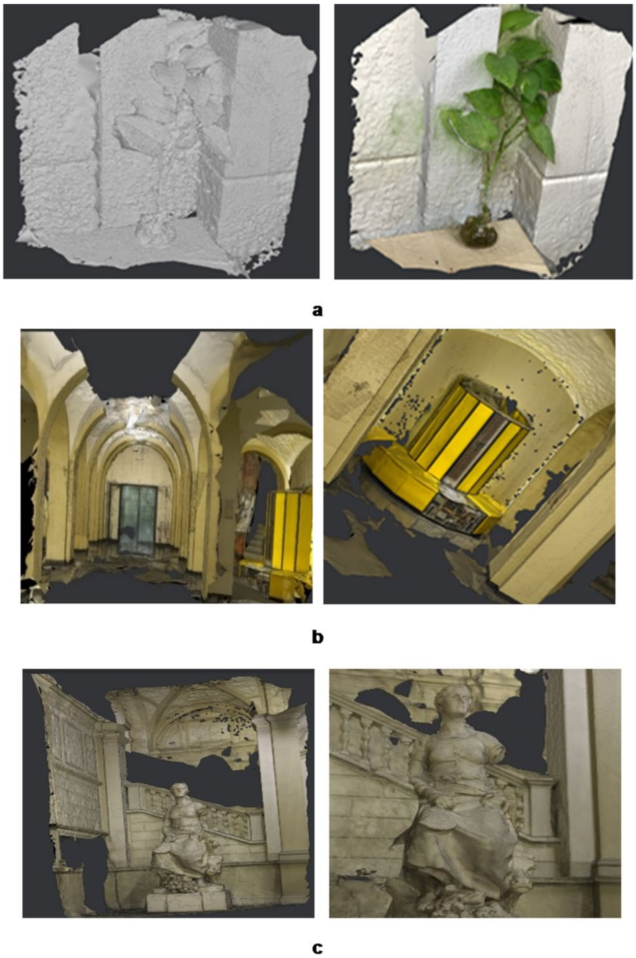

Fig. 3. (**a**) D Model, Mesh Model (Left), and Textured Model (Right) obtained from Money Plant Dataset (**b**) 3D Model of Old Computer Dataset - Front View (Left) and Zoomed View (Right). (**c**) 3D Model of Statue Dataset - Front View (Left) and Zoomed View (Right).

Employing the BRISQUE metric, we quantified the perceptual image quality of both our input dataset and the 3D model-generated images. The lower the BRISQUE score, the better the quality of the image, and scores up to 20 indicate good quality, and scores above 20 indicate not good quality[28]. The average BRISQUE score for the input dataset (Moneyplant) is computed at 30.32, as listed in Table 3, while the BRISQUE score for the images derived from the 3D model reached 20.85.

This comparison illustrates that the perceptual quality of the 3D model remains notably high, closely mirroring the characteristics of the original dataset. Consequently, the difference in BRISQUE scores is relatively low. Additionally, it is noteworthy that the initial images in the dataset, despite not being of optimal quality (considering the BRISQUE scores), still exhibit closely aligned scores.

Further, it is significant to observe that the standard deviation and coefficient of variation of the entropies and BRISQUE scores are not high, favoring the closely aligned datasets. Hence, once evaluated, the generated 3D model serves as a foundation for further processing, visualization, and analysis in various applications, such as computer vision, robotics, augmented reality, and virtual reality.

Table 4 shows that the 'Depth Map' calculations consume more time as compared to the other steps. The stepwise maximum time taken by this step is 52%, 49%, and 37% of the total time for the 'Money Plant' (21 images), 'Old Computer' (40 images), and 'Statue' (11 images) dataset, respectively. The step 'Camera Initialization' is the fastest. It is also observed that the 'Meshing' and 'Texturing' steps also take more time. We can observe from the rows that there is no trend like more the number of images more is the time-taken proportionally.

This suggests that the time taken also depends on imaging conditions, the number of features found, whether the images have more curved surfaces or more polygonal patches, etc., not just merely the number of images being processed. This analysis helps one to explore which steps need to be faster, maybe using faster algorithms, including parallel algorithms or appropriate data structures, to improve the speed of 3D model generation.

Since most of the steps include matrix operations and parallel algorithms[29] exist for these matrix operations. For example, for the matrix inversion algorithm proposed by Yu et al.[30] and for parallel matrix multiplication, the algorithm proposed by Sung et al.[31] can be used. For triangulation, the algorithm proposed by Yang et al.[32] may be used effectively.

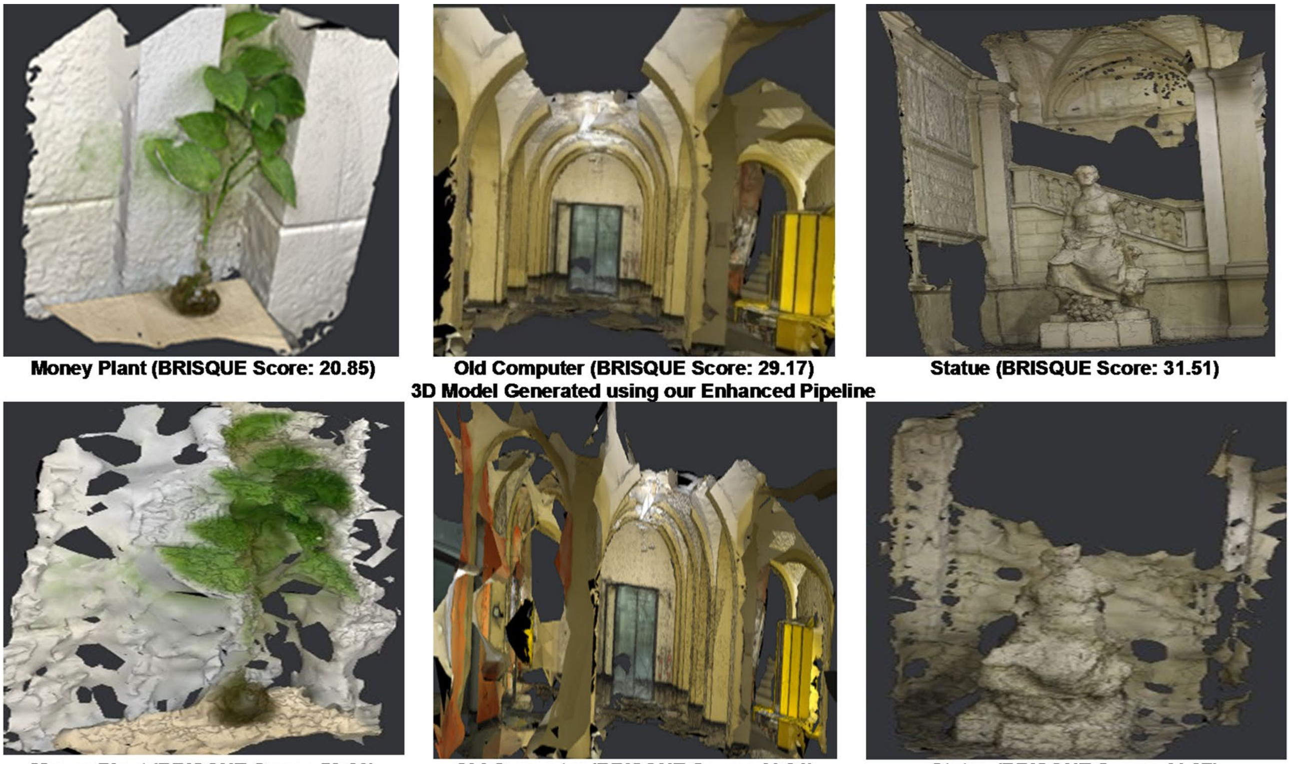

Fig. 4. 3D Model Generated Using standard photogrammetric pipeline with default settings.

Dataset	Statistics	Entropy	BRISQUE Score
Money Plant	Max	5.27	35.24
	Min	4.84	15.05
	Average	5.07	30.32
	Std. Dev.	0.01	5.21
	Coef. of Variation	0.02	0.17
Statue	Max	4.94	38.19
	Min	4.77	34.16
	Average	4.87	36.38
	Std. Dev.	0.04	1.20
	Coef. of Variation	0.01	0.03
Old Computer	Max	5.20	43.92
	Min	4.71	35.60
	Average	5.03	39.17
	Std. Dev.	0.11	2.14
	Coef. of Variation	0.02	0.05

Table 2. Statistics of the image entropy and image BRISQUE No-Reference image quality Scores.

Dataset	Entropy		BRISQUE Score	
	Maximum of Dataset	3D Model Image	Average of the Dataset	3D Model Image
Money Plant	5.27	4.72	30.32	20.85
Old Computer	5.20	4.70	36.38	29.17
Statue	4.94	4.30	39.19	31.51

Table 3. Comparison of evaluation Metrics.

Steps	Time Taken (s)		
	Proprietary Dataset	Public	
	Money Plant	Old Computer	Statue
Camera Initialization	0.51	0.14	0.14
Feature Extraction	70.00	109.00	173.00
Feature Matching	30.93	84.00	8.09
SFM	29.05	23.25	18.73
Prepare Dense Map	20.81	51.45	22.68
Depth Map	452.00	1110.00	367.00
Meshing	180.00	434.00	299.00
Texturing	90.00	463.00	102.00
Total Time taken	873.30	2274.84	990.64

Table 4. Comparison of time taken by each step of the 3D-Modeling Pipeline.

The No-Reference Image Quality Assessment using state-o-the-art metric BRISQUE is consistent under both distortions and noise[33,34]. For 3D-Model generation minimum two images (i.e. a stereo pair) is needed and considering a cuboidal volume or spherical volume six stereo-pair are sufficient and since usually the image captures from bottom to upward direction is not used, hence 11 images are sufficient[35]. So, in our experimentation we have not used a large dataset due to the following reasons: 1). Using a larger dataset is computationally extensive and does not add more details because usually around one thousand features distributed over entire region of an image are sufficient; 2). Larger dataset is used for applications where objective is to recognize objects, generate panoramic scenes, estimate pose, performing measurements in a dynamic environment; and 3). Since our objective is to generate 3D model of a single object with consistent image quality, a smaller dataset suffices for the same.

As we have seen from previous studies by Kim et al. (2017) and Boutsi et al. (2023), standard photogrammetric pipelines with default settings often result in inconsistent or suboptimal 3D reconstructions[36,37]. In our experiments, the improved pipeline consistently produced high-quality reconstructions across all three datasets, demonstrating more reliable and robust performance compared to default workflows.

Conclusions

In our research, we proposed some improvements in the bundle adjustment step and adaption of minimum triangulation angle and tiling buffer size in the photogrammetry pipeline to generate 3D models from a set of 2D images while thoroughly exploring various mathematical concepts essential at different stages of this process. We applied this slightly improved pipeline to a proprietary dataset (money plant) and two publicly available datasets (statue and old computer) to construct 3D models. The evaluation of the generated 3D model encompasses a comparative analysis utilizing metrics of image entropy, BRISQUE, and also visual perspective. It is observed that the 3D model derived from our own image dataset, despite its lower quality, exhibited improved results regarding perceptual quality, entropy, and BRISQUE scores. This shows the resilience and effectiveness of our approach, even when dealing with image datasets of suboptimal to low quality. Our multi-faceted evaluation, combining visual, entropy-based, and BRISQUE perspectives, consistently scores the high quality and fidelity of the generated 3D model. As practical applications, one can utilize the generated 3D model for further processing in visualization and augmented reality and to study about plant evolution and ecology. In the future, in addition to BRISQUE scores, other no-reference image quality assessment metrics may be used for comparison. As possible future work, we also suggest using parallel algorithms to experiment on larger datasets to make some of the steps, especially those involving matrix operations, faster. Also in future, we aim to assess the quality of the reconstructed 3D models through geometric evaluation and by testing the pipeline under diverse imaging conditions.

Data availability

The data is available with Trishna Paul and can be provided upon request.

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References

1. Iglhaut, J. et al. Structure from Motion Photogrammetry in Forestry: a Review. *Current Forestry Reports* **5** (3), 155–168. (Springer International Publishing, 2019). https://doi.org/10.1007/s40725-019-00094-3
2. Schenk, T. & Quarter, A. *Introduction To Photogrammetry* (The Ohio State University, 2005).
3. Mikita, T., Balková, M., Bajer, A., Cibulka, M. & Patočka, Z. Comparison of different remote sensing methods for 3d modeling of small rock outcrops. *Sens. (Switzerland)*. **20** (6). https://doi.org/10.3390/s20061663 (Mar. 2020).
4. Struck, R., Cordoni, S., Aliotta, S., Pérez-Pachón, L. & Gröning, F. Application of photogrammetry in biomedical science. in *Advances in Experimental Medicine and Biology*, vol. 1120, Springer New York LLC, 121–130. DOI: https://doi.org/10.1007/978-3-030-06070-1_10. (2019).
5. Verykokou, S. & Ioannidis, C. An Overview on Image-Based and Scanner-Based 3D Modeling Technologies. *Sensors* **23** (2), (2023). https://doi.org/10.3390/s23020596

6. Jasińska, A., Pyka, K., Pastucha, E. & Midtiby, H. S. A simple way to reduce 3D model deformation in smartphone photogrammetry. *Sensors* **23** (2). https://doi.org/10.3390/s23020728 (2023).
7. Leménager, M., Burkiewicz, J., Schoen, D. J. & Joly, S. Studying flowers in 3D using photogrammetry. *New Phytologist* **237** (5), 1922–1933, https://doi.org/10.1111/nph.18553 (2023).
8. Gallego, J., Beacco, A. & Slater, M. 3D objects reconstruction using frontal images. an example with guitars. in *Proceedings - International Conference on Image Processing, ICIP*, IEEE Computer Society, pp. 1291–1295. (2022). https://doi.org/10.1109/ICIP46576.2022.9897422
9. de Oliveira, A. et al. Projection of realistic three-dimensional photogrammetry models using stereoscopic display: A technical note. *Anat. Sci. Educ.* https://doi.org/10.1002/ase.2329 (2023).
10. Le, S. et al. Photogrammetric 3D reconstruction of Apollo 17 station 6: from boulders to lunar rock samples integrated into virtual reality. *Planet. Space Sci.* **240** https://doi.org/10.1016/j.pss.2023.105813 (2024).
11. Liu, W. et al. 3D Building model generation from MLS point cloud and 3D mesh using multi-source data fusion. *Int. J. Appl. Earth Obs. Geoinf.* **116**, 103171. https://doi.org/10.1016/j.jag.2022.103171 (2023).
12. Enesi, I. & Kuqi, A. Performance analysis for 3D reconstruction objects in meshroom and Agisoft—A comparative study. *Int. J. Online Biomedical Eng. (iJOE).* **19** (05), 22–41. https://doi.org/10.3991/ijoe.v19i05.37257 (2023).
13. Lau, B. K., Ariff, M. F. M. & Razali, A. F. Comparison between AliceVision meshroom and Pix4Dmapper software in generating Three-Dimensional (3D) photogrammetry software. *J. Adv. Geospatial Sci. Technol.* **2** (2), 117–138 (2022).
14. Aati, S., Rupnik, E. & Nejim, S. Comparative study of photogrammetry software in industrial field. *revue Française de photogrammétrie et de télédétection* **1** (221) 37–48, (2020). https://doi.org/10.52638/rfpt.2019.439
15. Westoby, M. J., Brasington, J., Glasser, N. F., Hambrey, M. J. & Reynolds, J. M. Structure-from-Motion' photogrammetry: A low-cost, effective tool for geoscience applications. *Geomorphology* **179**, 300–314. https://doi.org/10.1016/j.geomorph.2012.08.021 (2012).
16. Matys, M., Krajcovic, M. & Gabajova, G. Creating 3D models of transportation vehicles using photogrammetry. in *Transportation Research Procedia*, Elsevier B.V., pp. 584–591. (2021). https://doi.org/10.1016/j.trpro.2021.07.025
17. Shervais, K. Structure from Motion (SfM) Photogrammetry Field Methods Manual for Students Introduction to SfM for Field Education,June 09, *SfM Field Methods Manual* [online], Available: Earthscope Consortium. (2023).
18. Fu, Y. et al. Seamless texture optimization for RGB-D reconstruction. *IEEE Trans. Vis. Comput. Graph.* **29** (3), 1845–1859. https://doi.org/10.1109/TVCG.2021.3134105 (2023).
19. Lowe, D.G. Object recognition from local scale-invariant features. *Proc. Seventh IEEE Int. Conf. Comput. Vis.* **2**, 1150–1157. https://doi.org/10.1109/ICCV.1999.790410 (1999).
20. Computer Vision and Geometry Group of ETH & Zurich. Accessed on Jan 05 (2024). https://www.eth3d.net/datasets.
21. Elias, R. Projective Geometry for 3D Modeling of Objects, in 3D Surface Geometry and Reconstruction: Developing Concepts and Applications, U. Pati (ed.), IGI Global, ISBN-10: 1466601132, ISBN-13: 978-1466601130, ch. 2, pp. 28–48, Hershey, PA, (2012). https://doi.org/10.4018/978-1-4666-3994-2.ch008
22. Hartley, R. & Zisserman, A. Multiple View Geometry in Computer Vision. New York, NY, USA: Cambridge University Press. 0521540518 ISBN https://doi.org/10.1017/CBO9780511811685 (2003).
23. Loop, C. & Zhang, Z. Computing rectifying homographies for stereo vision. Proceedings. *IEEE Computer Society Conference on Computer Vision and Pattern Recognition* (Cat. No PR00149), Fort Collins, CO, USA, 1999, pp. 125–131 Vol. 1, (1999). https://doi.org/10.1109/CVPR.1999.786928
24. Forstner, W. & Wrobel, B. P. *Photogrammetric Computer Vision: Statistics, Geometry, Orientation and Reconstruction* 1st edn (Springer Publishing Company, 2023). Incorporate. ISBN978-3319115498.
25. Mello Román, J. C. et al. García torres. Entropy and contrast enhancement of infrared thermal images using the multiscale Top-Hat transform. *Entropy* **21** (3), 244. https://doi.org/10.3390/e21030244 (2019).
26. Chen, X. et al. No-reference color image quality assessment: from entropy to perceptual quality. *J. Image Video Proc.* **77** (2019). https://doi.org/10.1186/s13640-019-0479-7 (2019).
27. Liu, Q., Su, H., Duanmu, Z., Liu, W. & Wang, Z. Perceptual quality assessment of colored 3D point clouds. *IEEE Trans. Vis. Comput. Graph.* **29**, 8 (2023). Online publication date: 1-Aug-2023.
28. Dr, S. et al. Image Quality Assessment Based On NIQE, PIQE, GLCM, and SVM. *International Journal of Emerging Technologies and Innovative Research (www.jetir.org)*, ISSN: 2349–5162, **6** (4), 159–163, (2019).
29. Grama, A., Gupta, A., Karypis, G. & Kumar, V. Introduction to Parallel Computing, Second Edition, Addison Wesley, ISBN: 0-201-64865-2, (2003).
30. Yu, D., He, S., Huang, Y., Yu, G. & Yang, L. A fast parallel matrix inversion algorithm based on heterogeneous multicore architectures. *2015 IEEE Global Conference on Signal and Information Processing (GlobalSIP)*, Orlando, FL, USA, pp. 903–907, (2015). https://doi.org/10.1109/GlobalSIP.2015.7418328
31. Bae, S. E., Shinn, T. W. & Takaoka, T. A faster parallel algorithm for matrix multiplication on a mesh array. *procedia computer science,* **29**, Pages 2230–2240, ISSN 1877 – 0509, (2014). https://doi.org/10.1016/j.procs.2014.05.208
32. Yang, H. et al. PatCC1: an efficient parallel triangulation algorithm for spherical and planar grids with commonality and parallel consistency. *Geosci. Model. Dev.* **12**, 3311–3328. https://doi.org/10.5194/gmd-12-3311-2019 (2019).
33. Mirza, M. S., Gala, A., Devarakota, P., Mantini, P. & Shah, S. K. Integrating image quality assessment metrics for enhanced segmentation performance in reconstructed imaging datasets. *Proceedings Copyright.* **450**, 457 (2025).
34. Pennada, S., Perry, M., McAlorum, J., Dow, H. & Dobie, G. Threshold-based BRISQUE-assisted deep learning for enhancing crack detection in concrete structures. *J. Imaging.* **9** (10), 218 (2023).
35. Stojaković, V. Terrestrial photogrammetry and application to modeling architectural objects. *Facta Universitatis-Series: Archit. Civil Eng.* **6** (1), 113–125 (2008).
36. Kim, H. S. & Park, C. J. High quality texture reconstruction from multi-view images. In *2017, International Conference on Information and Communication Technology Convergence (ICTC),* pages 1112–1114, (2017). https://doi.org/10.1109/ICTC.2017.8190870
37. Boutsi, A. M., Ioannidis, C. & Verykokou, S. Multi-resolution 3d rendering for high-performance web ar. *Sensors* **23** (15), 6885 (2023).

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